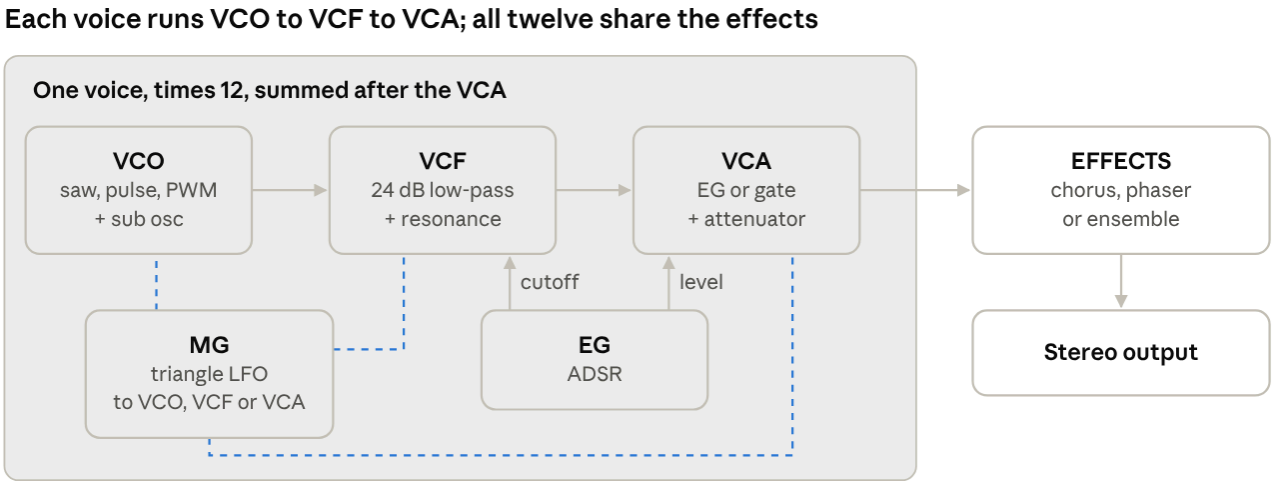
Polytwelve User Manual

Oct 14, 2026 · @David

Introduction

Polytwelve is a programmable 12-voice analog-style synthesizer by Ticco Ross (2026), version 0.6.0. Each voice has one oscillator with a sub-oscillator, a resonant low-pass filter (Classic ladder or SSM2044 model), an envelope, an amplifier and its own modulation generator (MG). The twelve voices are summed and pass through a stereo effects section (Chorus, Phaser or Ensemble).



signal flow · one voice of twelve, then the shared effects

Solid arrows carry the sound and the envelope; the dashed lines are the MG, which reaches one destination at a time, chosen by its Mode switch.

It plays three ways: **Poly** (twelve notes at once), **Unison** (all twelve voices on one note) and **Chord** (one key plays a stored chord). A built-in arpeggiator, a preset browser with favourites and per-voice analog drift complete it.

Formats

System	Formats	Installer
macOS 10.13 or later (Apple Silicon and Intel)	Audio Unit, VST3, standalone app	`Polytwelve-0.6.0-macOS.pkg`
Windows 10 or later (64-bit)	VST3, standalone app	`Polytwelve-0.6.0-Windows-Setup.exe`

System	Formats	Installer
Linux x86_64	VST3, LV2, standalone app	`Polytwelve-0.6.0-Linux-x86_64.tar.gz`

Installing

- **macOS:** double-click the `.pkg` and follow the steps. Click **Customize** to skip the AU, VST3 or app. The first time, macOS may block it: open **System Settings → Privacy & Security** and click **Open Anyway**.
- **Windows:** run the Setup `.exe`. If SmartScreen appears, click **More info → Run anyway**. The VST3 goes to `C:\Program Files\Common Files\VST3`.
- **Linux:** unpack the archive and run `./install.sh`. It installs for your user to `~/.vst3`, `~/.lv2` and `~/.local/bin`, with no root needed.

Afterwards, rescan plug-ins in your DAW. In Ableton Live on a Mac: **Settings → Plug-Ins**, turn on **Use Audio Units v2**, hold **Option** and click **Rescan**. Polytwelve appears under **Ticco Ross**.

Panel overview

The panel has a header with the preset bar and three rows of sections, laid out in the order the sound travels.

Row	Sections, left to right
Header	Name and version (e.g. v0.5.0) · preset bar: `<` · preset name · `>` · ☆ favourite · SAVE · DEL · ★ ONLY
1	MASTER · VCO · VCF · VCA
2	EG · MG · VOICE
3	ARPEGGIATOR · EFFECTS · WHEELS · CHORD MEMORY

Working the controls:

- **Knobs:** drag up or down. Hold the mouse on a knob to see its exact value (Hz, ms, dB, semitones). **Double-click** returns it to its default.
- **Selectors:** click and pick from the list.
- **White switches:** click to toggle. The red LED shows the switch is on.

- Every control can be automated from your DAW and is saved with your project and with presets.

Master and VCO

MASTER

Control	Range	What it does
Volume	0–10 (default 7.5)	Overall output level, after the effects.
Tune	±100 cents	Fine-tunes all voices by up to one semitone either way.

VCO (oscillator)

Each voice has one oscillator plus a sub-oscillator. The waveforms are band-limited, so high notes stay clean.

Control	Range	What it does
Octave	16′ · 8′ · 4′	Shifts the oscillator down one octave, normal, or up one octave.
Waveform	Saw · Pulse · PWM	**Saw:** bright and full. **Pulse:** hollow; set its width with PW/PWM. **PWM:** a pulse whose width sweeps on its own, for a moving, chorused tone.
PW/PWM	0–10	In **Pulse** mode, the pulse width: 0 = square (50%), 10 = thin (95%). In **PWM** mode, how far the width sweeps. No effect on Saw.
PWM Speed	0.1–12.8 Hz	How fast the width sweeps in PWM mode.
Sub Osc	Off · 1 Oct · 2 Oct	Adds a square wave one or two octaves below the main oscillator, at equal level. Locked to the oscillator, like the original Polysix divider.

Tip: Pulse with the sub at 1 Oct and a low cutoff gives classic Polysix basses. PWM with Ensemble gives lush strings.

VCF, VCA and EG

VCF (filter)

A resonant low-pass filter with two characters, chosen with **TYPE**. Both self-oscillate at high resonance and track the keyboard in tune.

Control	Range	What it does
Type	Classic · SSM2044	**Classic:** a 4-pole (24 dB/oct) ladder with soft saturation in its feedback loop; deep, round resonance. **SSM2044:** a model of the Korg Polysix filter chip; more open, gentler resonance, thinner bass at high resonance.
Cutoff	27.5 Hz – 17 kHz	The filter's brightness. Lower = darker.
Resonance	0–10	Emphasis at the cutoff. Near 10 the filter sings on its own (from about 8.6 on SSM2044).
EG Intensity	−10 to +10	How far the envelope moves the cutoff: up to ±5 octaves. Negative values sweep it down instead of up.
Kbd Track	0–10	How much the cutoff follows the keys. At 10 it tracks the keyboard one-to-one, centred on C3.

About the SSM2044 model. It follows the Polysix filter circuit: four OTA stages where the first uses a 560 pF capacitor and the other three 6.8 nF, which puts the first pole about 12× higher. The slope is therefore about 18 dB/oct just above cutoff and 24 dB/oct further up. The small input level from the circuit's 4.7 k / 220 Ω divider gives a soft saturation in every stage, and the global resonance feedback thins the bass as resonance rises, the classic Polysix trait. At the same Cutoff setting it sounds a little darker than Classic, so open Cutoff slightly when switching.

VCA (amplifier)

Control	Range	What it does
Mode	EG · Gate	**EG:** volume follows the envelope. **Gate:** volume is fully on while a key is held and off when released, organ-style (the EG still drives the filter).
Attenuator	±10 dB	Level of each voice before the effects. Lower it if high resonance or Unison gets too loud.

EG (envelope)

One ADSR per voice, shared by the filter (through EG Intensity) and the amplifier. Its curves are exponential, like an analog envelope.

Control	Range	What it does
Attack	1 ms – 10 s	Time to rise to full level.
Decay	1 ms – 10 s	Time to fall to the sustain level.
Sustain	0–10	Level held while the key is down.
Release	1 ms – 10 s	Time to fade after the key is released.

The time knobs are logarithmic: 2.5 ≈ 10 ms, 5 ≈ 100 ms, 7.5 ≈ 1 s, 10 = 10 s. With **Analog** turned up, each voice's decay and release differ slightly.

MG (modulation generator)

The MG is a triangle LFO, one per voice, that fades in after each note starts and modulates one destination. With **Analog** turned up, each voice's MG runs at a slightly different rate.

Control	Range	What it does
Frequency	0.01–20 Hz	Speed. 2.5 ≈ 1 Hz, 5 ≈ 8 Hz, 7.5 ≈ 15 Hz.
Delay	1 ms – 3.2 s	How long the modulation takes to fade in after a note starts. Restarts with every note.
Level	0–10	Depth. The curve is gentle at the bottom for fine vibrato.

Control	Range	What it does
Mode	VCO · VCF · VCA	**VCO:** vibrato (up to ±12 semitones at full Level). **VCF:** filter wobble (up to ±4 octaves). **VCA:** tremolo.

The **mod wheel** always adds vibrato from the same triangle, up to ±2 semitones, whatever the Mode, Level or Delay settings.

Tip: Mode VCO, Delay 6, Level 3 and Frequency 4 gives a delayed vibrato for leads.

Voice

The Voice section sets how the twelve voices are played and how "analog" they behave.

Control	Range	What it does
Mode	Poly · Unison · Chord	How keys are assigned to voices (below).
MEMORY	button	Stores the chord you are holding and switches to Chord mode. See **Chord Memory**.
Hold	on / off	Latches notes: they keep sounding after you let go. Turning Hold off releases them. A sustain pedal does the same.
Glide	Off, 5 ms – 3 s	Portamento: each new note slides in pitch from the previous one, taking about this long. 0 = off.
Legato	on / off	When on, notes glide only if you press the new key before releasing the old one; detached notes jump straight to pitch.
Analog	0–10	Amount of per-voice variation: a small fixed tuning offset and slow pitch drift, plus differences in cutoff, decay, release and MG and PWM rates. 0 = perfectly stable.
Detune	0–10	A fixed tuning offset per voice, up to ±25 cents. Most audible in Unison.

Voice modes

- **Poly:** up to twelve notes at once. Each new note takes the next free voice in turn (round-robin), which lets release tails ring. With all twelve busy, the oldest released voice is reused first, then the oldest held one.

- **Unison:** all twelve voices play one note, stacked with their Detune and Analog offsets, for a huge mono sound. The last key pressed wins; releasing it returns to a key still held.
- **Chord:** every key plays the stored chord, transposed. One chord sounds at a time, with the same last-key behaviour as Unison.

Switching modes while playing releases the sounding notes and restarts any held keys in the new mode.

Glide (portamento)

With **Glide** above 0, each new note slides up or down from the previous pitch. It works in every mode and with the arpeggiator:

- **Poly:** the new note glides from the last note you played.
- **Unison and Chord:** every voice glides from where it is, so whole chords slide together.
- **Legato on:** glide happens only between overlapping notes (press the next key before letting go of the last). Play detached for jumps, connected for slides.

The Square Lead preset uses a short legato glide.

Chord Memory

Chord Memory stores a chord of up to twelve notes; after that, every single key plays the whole chord, transposed to that key. It is the classic Polysix trick for parallel stabs and pads.

Storing a chord

1. Hold the chord on your MIDI keyboard, e.g. A2, C3, E3, G3.
2. While holding it, press **MEMORY** in the Voice section.
3. Let go. Mode is now **Chord** and the MEMORY LED is lit. Pressing D3 plays Dm7, F3 plays Fm7, and so on.

Other ways it works:

- **MEMORY with no keys held** switches to Chord mode using the chord already stored.
- **Switching Mode to Chord while holding two or more keys** also stores them.
- A **single held key** stored with MEMORY plays just that note, which is useful for octave or unison layers.
- The chord is saved with presets and with your project. The default chord is a minor triad (C3 · Eb3 · G3).
- Chord mode works with the arpeggiator: each step plays the whole chord.

The CHORD MEMORY display

The amber read-out in the bottom row shows the stored chord:

- **Top line:** the chord name, e.g. Am7, Cmaj7, Fsus4, or C/E for an inversion. It recognises triads, sus, power chords, 6ths, 7ths, 9ths and 11ths; other shapes show as "4-note chord".
- **Bottom line:** the notes as stored, lowest first, e.g. A2 · C3 · E3 · G3. Note names follow Ableton's convention (middle C = C3).

The display glows in Chord mode and dims in Poly or Unison, where the chord is kept but not used.

Arpeggiator

The arpeggiator plays the keys you hold one at a time, in pitch order, repeated over a range of octaves. Each step sounds for half its length.

Control	Range	What it does
Arpeggio	on / off	Turns the arpeggiator on.
Speed	0.5–24 steps per second	Rate when Sync is off.
Range	1 Oct · 2 Oct · Full	How many octaves the pattern covers: one, two, or four.
Mode	Up · Down · Up/Down	Direction. Up/Down doesn't repeat the top and bottom notes.
Latch	on / off	Keeps the pattern running after you let go. The next chord you play replaces it.
Sync	on / off	Follows your DAW's tempo instead of the Speed knob.
Sync Rate	1/4 · 1/8 · 1/8T · 1/16 · 1/16T · 1/32	Step length when Sync is on.

With Sync on and the DAW playing, steps lock to the song's beat grid. With the transport stopped, it runs at the project tempo.

The arpeggiator feeds the voice modes: in **Poly** the steps overlap through their release, in **Unison** each step is the full twelve-voice stack, and in **Chord** each step plays the stored chord.

Effects

One effect at a time processes all twelve voices together and makes the output stereo. A single knob sets its speed or intensity.

Mode	Speed/Int knob controls	Sound
Off	—	Dry, mono in the centre.
Chorus	Speed, 0.1–4 Hz	A modulated delay, opposite on left and right. Widens leads and pads.
Phaser	Speed, 0.05–6.4 Hz	A six-stage phaser with feedback, its sweep offset between left and right.
Ensemble	Intensity	Three modulated delay lines with slow and fast movement, for the lush string-machine sound.

The chorus and ensemble wet signals are gently rolled off above about 9 kHz, like the bucket-brigade circuits they imitate. Switching modes resets the effect, so change it between notes.

Wheels

The WHEELS section, left of the chord display, has an on-screen pitch-bend wheel and modulation wheel for playing with the mouse.

Control	Range	What it does
Bend Range	0–12 semitones (default 2)	How far the pitch wheel bends, up and down.
PITCH	up / down	Drag to bend all voices. Springs back to the centre when you let go.
MOD	0–max	Drag up to add MG vibrato, up to ±2 semitones. Stays where you leave it.

The wheels also move on their own when your MIDI keyboard sends pitch bend or mod wheel, so they always show the current position.

Presets

A preset stores every control plus the stored chord. The preset bar sits at the top right of the panel.

Control	What it does
`<` `>`	Previous / next preset.
Preset name	Opens the browser: **Favourites**, **Factory** and **User** lists, plus Save As…, Open Preset File… and Show Preset Folder. Favourites are marked ★.
☆ / ★	Adds or removes the current preset from favourites. Turns yellow when it is one.
SAVE	Saves the current sound as a user preset. It asks for a name, and asks before replacing an existing one.
DEL	Deletes the current user preset, after confirming. Not available for factory presets.
★ ONLY	When lit, `<` and `>` step through favourites only.

Your project remembers which preset was loaded. Factory presets also appear as programs in hosts that show a program list.

Factory presets

Preset	Character
Init	Plain saw, the starting point
Strings	Slow saw pad through the Ensemble
Brass	Swelling filter envelope
Bass	16′ pulse with sub, resonant pluck
Square Lead	Unison square with legato glide, delayed vibrato and chorus
Rezzy Arp	Resonant saw on a synced 1/16 arpeggio, through the phaser
Filter Sweep	PWM with the MG slowly sweeping the filter

Preset	Character
SSM Bass	16′ saw with sub through the SSM2044, resonant and plucky
SSM Brass	Polysix-style brass on the SSM2044 with the Ensemble
Minor Chord	Chord mode with a minor triad, plucky, with chorus

Preset files

User presets are `.p6preset` files, one per sound, which you can copy, back up or share. Favourites are listed in `favourites.txt` in the same folder.

System	Preset folder
macOS	`~/Library/Audio/Presets/Ticco Ross/Polytwelve`
Windows	`Documents\Ticco Ross\Polytwelve\Presets`
Linux	`~/.local/share/Ticco Ross/Polytwelve/Presets`

To add a preset someone sent you, use **Open Preset File…** and then SAVE it, or drop the file into the folder.

MIDI implementation

Polytwelve responds on all MIDI channels.

Message	Effect
Note on / off	Plays notes. Velocity is ignored, as on the original Polysix.
Pitch bend	Bends all voices by up to the **Bend Range**; the on-screen PITCH wheel follows it.
CC 1 (mod wheel)	Adds MG vibrato, up to ±2 semitones; the on-screen MOD wheel follows it.
CC 64 (sustain pedal)	Works like **Hold**: notes keep sounding until the pedal is released.

Message	Effect
CC 120 / CC 123	All sound off / all notes off: stops every voice and clears held notes.

All 38 controls are exposed to the host for automation and MIDI mapping. In Ableton Live, use the plug-in's **Configure** button to add them to the device, then map them like any Live control.

Specifications and credits

Item	Value
Version	0.6.0 (shown next to the name at the top of the panel)
Polyphony	12 voices
Per voice	1 oscillator (saw, pulse, PWM) + sub-oscillator, resonant low-pass filter (Classic 24 dB/oct ladder or SSM2044 model), ADSR, VCA, triangle MG, glide
Effects	Chorus, Phaser, Ensemble (stereo)
Output	Stereo
Formats	AU (macOS), VST3, LV2 (Linux), standalone app
Parameters	38, all automatable

Uninstalling

- **macOS:** delete `Polytwelve.component` from `/Library/Audio/Plug-Ins/Components`, `Polytwelve.vst3` from `/Library/Audio/Plug-Ins/VST3` and `Polytwelve.app` from Applications.
- **Windows:** Settings → Apps → Polytwelve → Uninstall.
- **Linux:** run `./uninstall.sh` from the downloaded folder.

Uninstalling leaves your presets in place.

Credits

Polytwelve is designed by Ticco Ross, 2026.